Food Security Drivers in Smallholder Farming Households

A Multi-Country Analysis Using the RHoMIS Dataset (54,873 households, 35 countries).

Sector: Agriculture
Team: Group 7 (Section C)
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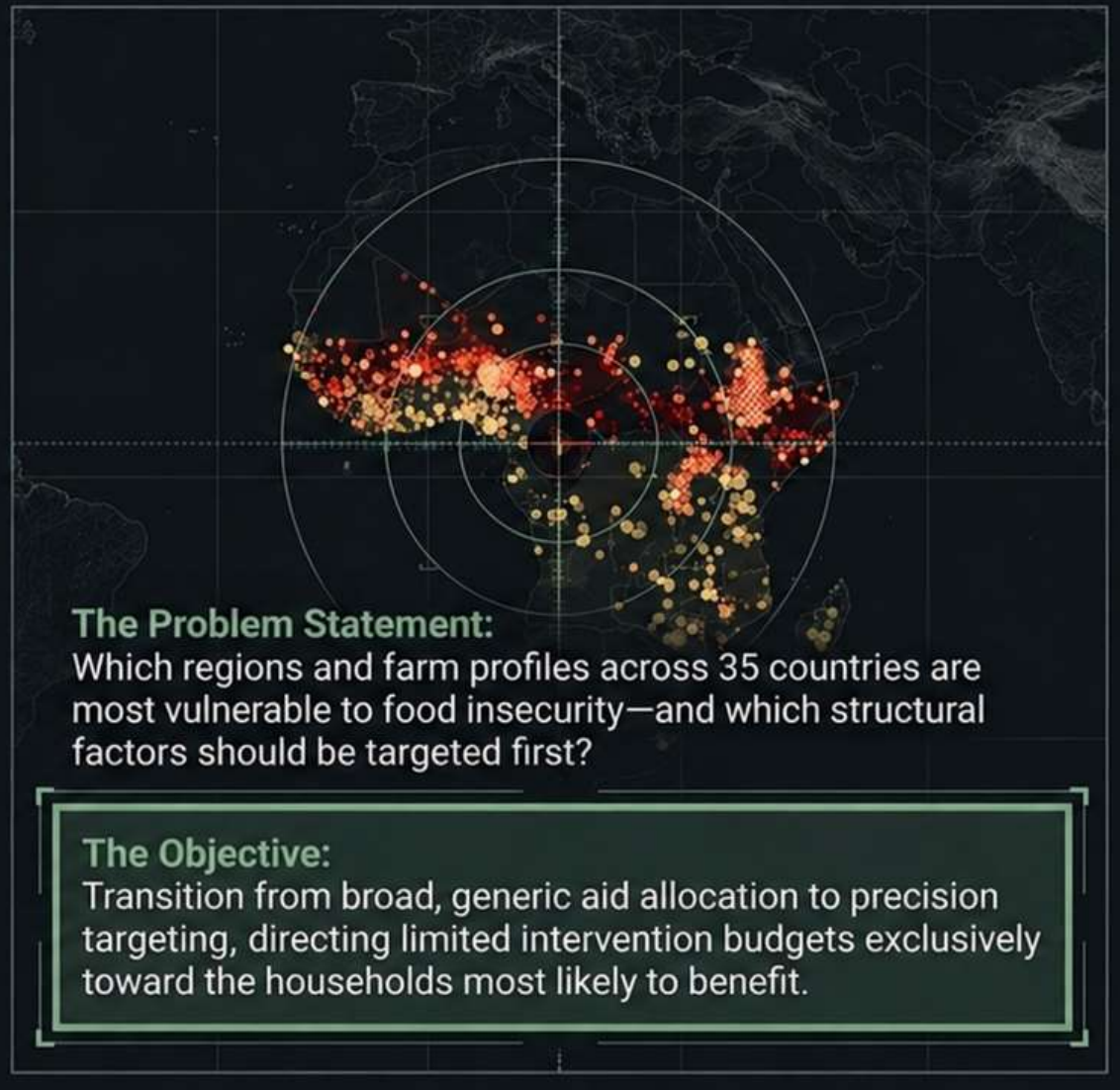
Interventions fail when resources are spread evenly across uneven risks

THE STATUS QUO

Sector Context:
Smallholder agriculture supports
500 million rural families globally.
However, development organisations and policymakers lack clear evidence on which specific farm profiles and regions carry the highest risk.

THE OBJECTIVE



The Problem Statement:
Which regions and farm profiles across 35 countries are most vulnerable to food insecurity—and which structural factors should be targeted first?

The Objective:
Transition from broad, generic aid allocation to precision targeting, directing limited intervention budgets exclusively toward the households most likely to benefit.

Distilling 1,599 raw survey variables into a 46-column analytics layer.

Source Data: RHoMIS Dataset
(54,873 rows × 1,599 columns, 2015–2023)

Refinery Pipeline

Pipeline Chamber 1: Standardizing Units

Converted local land units (bigha, cuerda) into hectares using a strict 28-unit mapping table. No zero-filling or data invention applied to unmapped units.

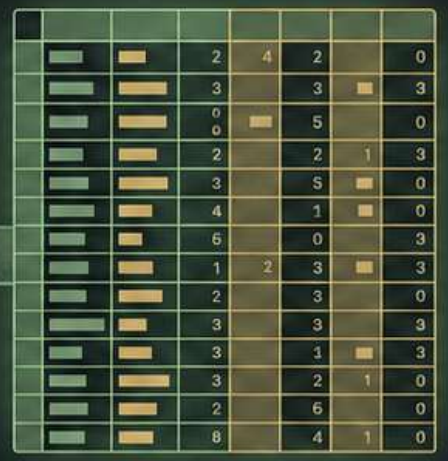
Pipeline Chamber 2: Rebuilding Household Size

The original count_people field was 49.4% missing. We engineered a new household_size variable by summing 8 specific age-band columns, yielding zero missing values.

Pipeline Chamber 3: Fixing Anomalies

Detected and corrected age fields where birth years were erroneously entered instead of actual ages, tying the correction to the specific survey year of each row.

46 Decision-Ready Columns



Seven metrics define the architecture of household vulnerability.

Executive Console

67.7% Global Food Shortage Rate

3.0 Months
Median duration of shortage

$112 PPP
Median Income per MAE

17.4pp

Irrigation Gap

72.4% shortage in rainfed farms vs. 55.1% in irrigated farms (only 20.2% have access).

29.1pp

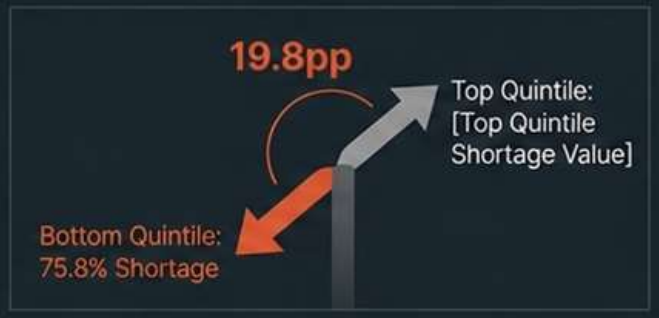
Productivity Gap

83% shortage in the lowest harvest quartile vs. 54% in the highest.

19.8pp

Income Gap

The shortage rate difference between the bottom quintile (75.8% shortage) and the top quintile.



Food insecurity is **structurally** concentrated, not geographically inevitable

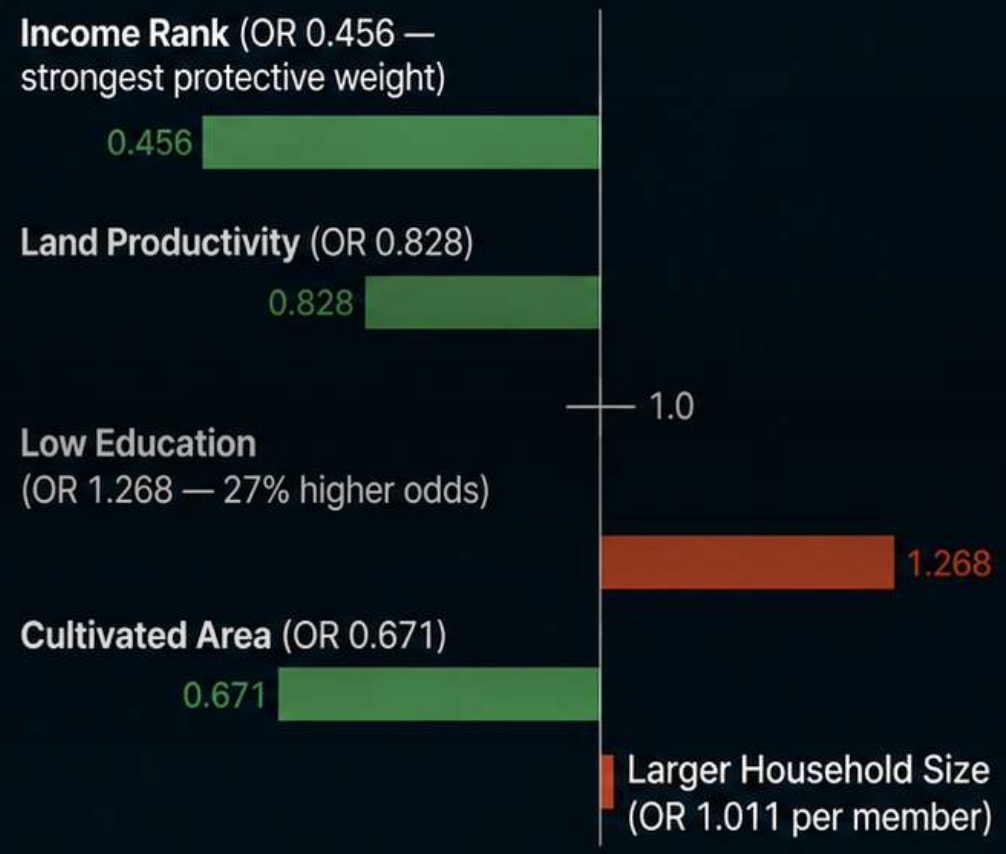
The Extremes of Risk:
While the global shortage rate is 67.7%, country-level variation proves that local infrastructure and climate are primary moderators.

Structural Success: India (10.5%) and Nepal (18.1%) demonstrate that smallholder farming does not guarantee food insecurity when supported by better irrigation and infrastructure frameworks.

Structural Failure: DRC (94.5%) and Rwanda (94.3%) exhibit near-total population shortage, compounded by low median incomes ($112 PPP globally).

Logistic regression and K-means clustering isolate the highest-risk profile.

The Regression Drivers (Odds Ratios)



The 14,330 Household Target Cluster

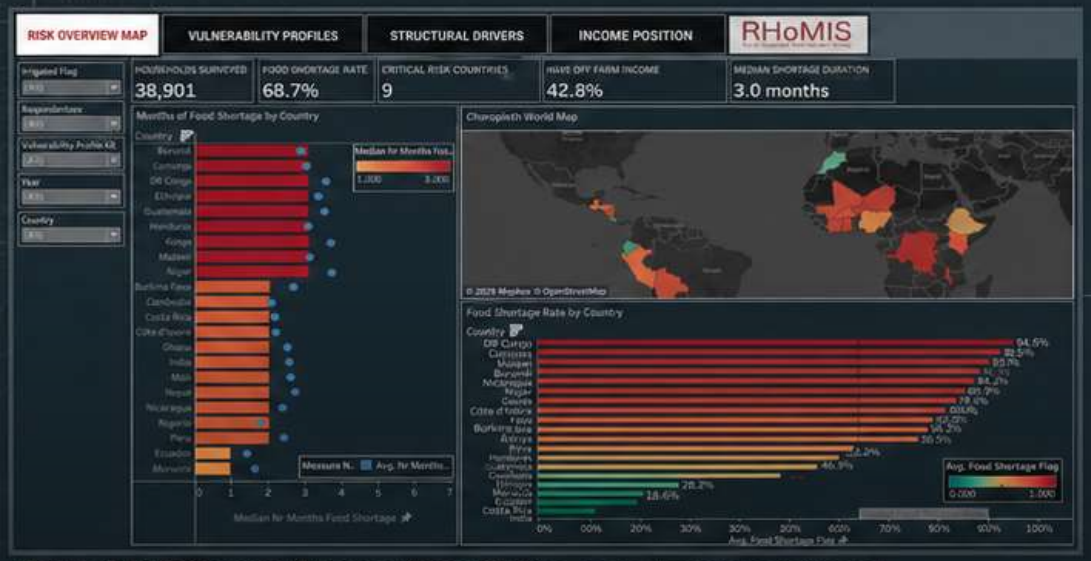
K-means clustering identified 4 distinct profiles. The undisputed target for immediate intervention is the **Low-income rainfed cluster.**

Cluster Specs:

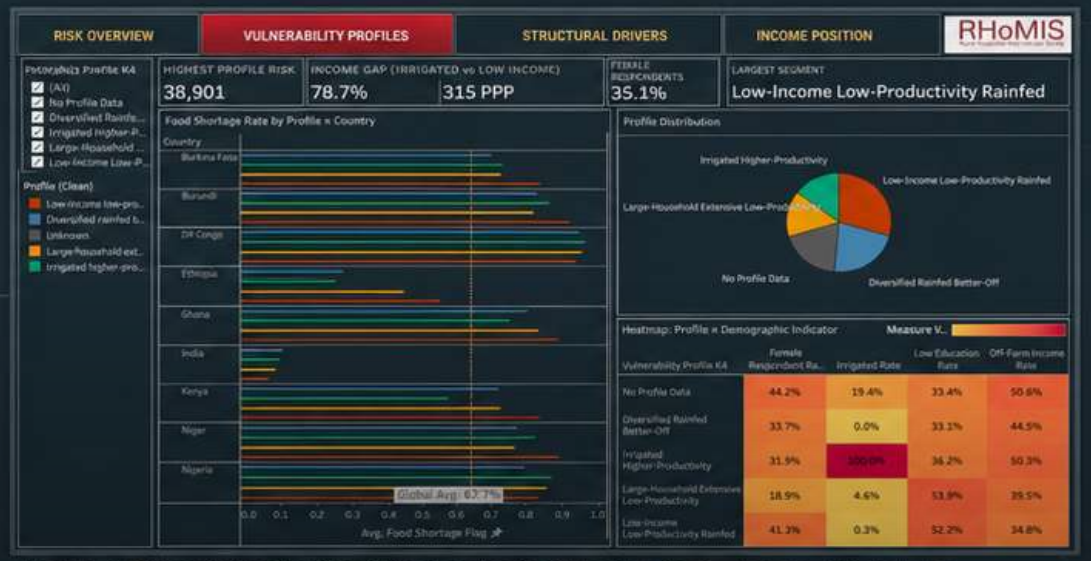
- 14,330 households.
- 78.7% shortage rate.
- Only 0.3% irrigation access.
- Lowest income rank (0.24).

An interactive diagnostic tool bridging macro-geography to micro-drivers.

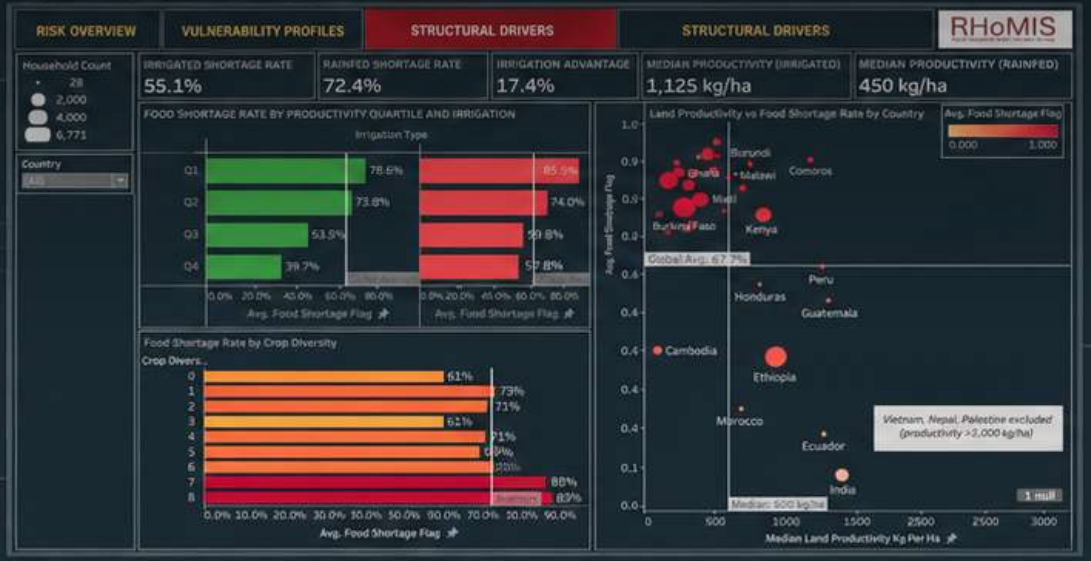
Tool allows seamless cross-dashboard navigation without returning to menus.



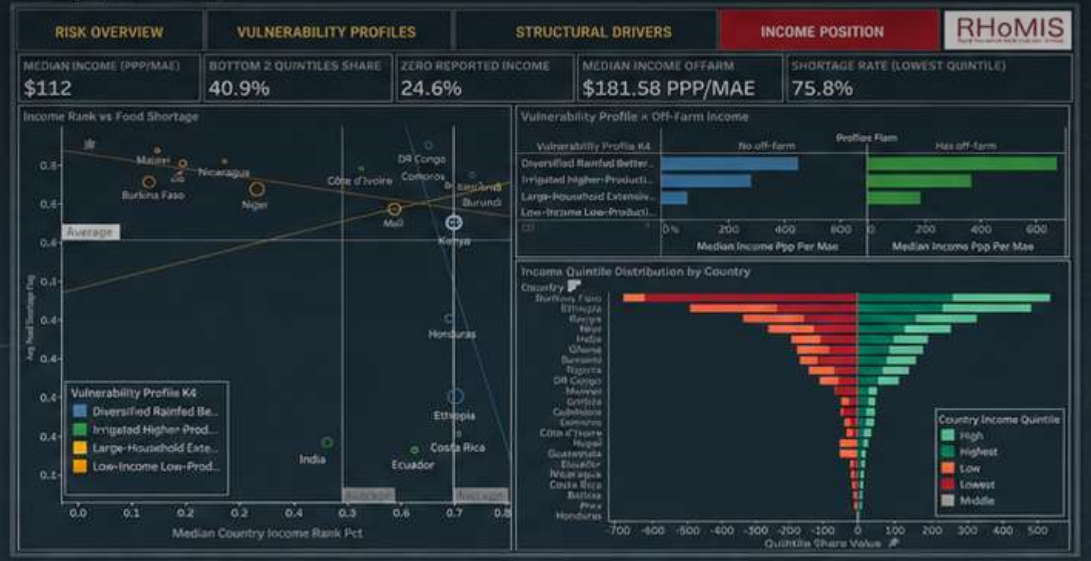
Module 1: Risk Overview (Where is it worst?): Maps country-level shortage rates and the critical 3-month median duratilen.



Module 2: Vulnerability Profiles (Who is suffering?): Isolates the low-income rainfed cluster demographic heatmaps.



Module 3: Structural Drivers (What is failing?): Cross-filters the 17.4pp irrigation advantage against the productivity quartiles.



Module 4: Income Position (What amplifies the risk?): Tracks the compounding danger of bottom-quintile income dependency and off-farm flags.

NotebookLM

PREMIUM ANALYTICS DASHBOARD

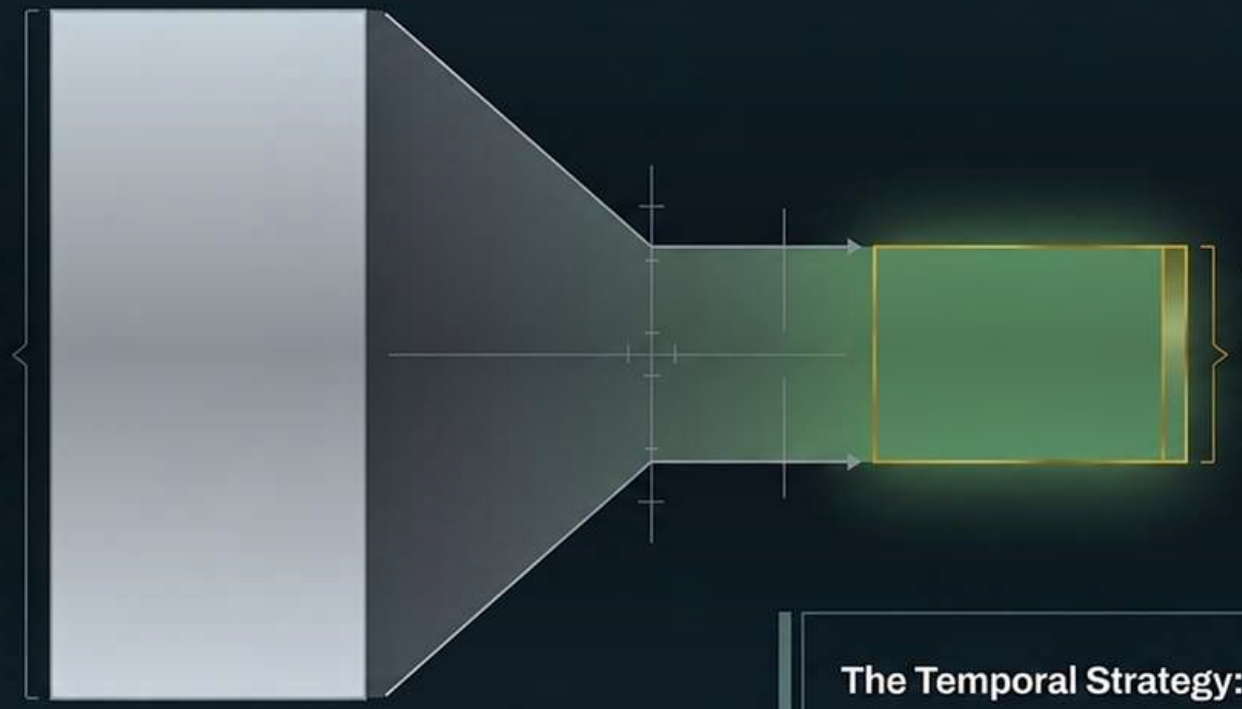
Three precision-targeted interventions for high-ROI development budgets.

Targeting Matrix	INSIGHT	RECOMMENDATION	EXPECTED IMPACT
Irrigation	Rainfed farms face a 17.4pp higher shortage rate. The highest-risk K-means cluster has near-zero (0.3%) irrigation access.	Prioritise small-scale/drip irrigation exclusively for the 14,330 HH 'low-income rainfed' cluster in DRC and Rwanda.	**Move households completely out of acute insecurity.**
Productivity	A massive 29.1pp gap exists between productivity quartiles. Extensive land without productivity yields a 72.3% shortage rate.	Target the ~24,000 households in Q1 and Q2 with direct extension services (seed varieties, soil management).	**Estimated 10–15pp reduction in shortage rates for bottom-quartile farms.**
Income & Education	Income rank is the strongest predictor (OR 0.456), and low education severely amplifies risk (OR 1.268).	Bundle off-farm microenterprise pathways with adult literacy support specifically for bottom-quintile households (~10,200 HH).	**Estimated 4–8pp reduction in shortage risk per upward income step.**

Precision targeting increases measurable impact per household reached.

The Addressable Challenge: Exactly 32,097 households are currently trapped in food shortage.

Cost of Inaction:
Failing to target means development budgets are wasted on the 32.3% of households that do not require emergency intervention (spreading resources across all 47,000+ HHs)



The Efficiency Yield (ROI):
Irrigation targeting yields ~2,300 households exiting insecurity per 1,000 reached (leveraging the 17.4pp gap).

The Temporal Strategy:
Because the median shortage duration is exactly 3.0 months, interventions do not require year-round generic aid. Capital should be deployed for highly targeted, seasonal lean-period bridging and permanent structural upgrades.

Current analytical boundaries and the roadmap for systemic integration

Structural Limitations:

- **Cross-Sectional Constraints:** Data provides associative bounds, not causal proof. Year distribution is uneven (2018 = 22%, 2015 = 2%).
- **Data Artifacts:** Extreme income outliers (up to 350B PPP) required using within-country rank modeling.
- **Reporting Bias:** Food shortage remains self-reported, though statistically validated against FIES (p=0.587).

Next Steps for the Architecture:

- **Predictive Screening:** Deploy the 5-variable regression model as a lightweight predictive screening tool for incoming survey waves.
- **Geospatial Integration:** Link GPS coordinates to satellite NDVI and climate data to isolate farm structure from localized weather shocks.
- **Longitudinal Linkage:** Connect future survey waves to enable within-household causal estimates.

Team & Technical Contributions

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Note: GitHub repository contains full documentation, notebooks, and ETL pipelines validating these findings.